23. Number Codes (Set B)

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — A lock code:

A suitcase lock uses this code: 2 = B, 5 = R, 7 = D. What is the code for 257?

Answer: 2 = B, 5 = R, 7 = D, so 257 → B, R, D. Answer: **BRD**.

Example 2 — A two-step rule:

A code is formed by doubling a number and then adding 3. What is the code for 6?

Answer: 6 × 2 = 12, then 12 + 3 = **15**.

Example 3 — Working out the rule from a pair:

In a code, 2 → 9 and 4 → 17. The same rule is used for every number. What is the code for 6?

Answer: From 2 to 4, the number goes up by 2 and the code goes up by 8, so the rule multiplies by 4. Checking: 2 × 4 = 8, plus 1 = 9. ✓ So the rule is '×4, then +1'. For 6: 6 × 4 = 24, plus 1 = **25**.

Example 4 — Multiple-choice:

A code is formed by subtracting 2 from a number and then multiplying by 4. What is the code for 9? A) 24 B) 28 C) 32 D) 20

Answer: 9 − 2 = 7, then 7 × 4 = 28. Answer: **B) 28**.

Example 5 — A digit cipher:

In a code, every digit has 2 added to it, wrapping back round to 0 after 9. What is the code for 793?

Answer: 7 → 9, 9 → 1 (wraps round past 9), 3 → 5. Answer: **915**.

Foundation

15 questions

1. A key safe uses this code: 2 = B, 5 = R, 7 = D. What is the code for 257?

2. A bicycle lock uses this code: 1 = F, 4 = N, 6 = S. The code NFS stands for which number?

3. A locker uses this code: 2 = M, 4 = A, 9 = T. What is the code for 492? A) ATM B) AMT C) TAM D) MAT

4. A diary lock uses this code: 1 = J, 3 = O, 8 = Y. What is the code for 138?

5. A toy robot uses this code: 0 = P, 5 = I, 6 = G. The code PIG stands for which number?

6. A treehouse code uses: 2 = C, 4 = A, 7 = T. What is the code for 247?

7. A car alarm uses this code: 1 = B, 3 = U, 9 = S. The code BUS stands for which number?

8. A padlock uses this code: 0 = O, 5 = N, 6 = E. What is the code for 605?

9. A vending machine uses this code: 1 = H, 2 = E, 3 = N. The code HEN stands for which number?

10. A pirate's map uses this code: 0 = O, 4 = G, 7 = D. What is the code for 704?

11. A school gate uses this code: 3 = A, 6 = N, 8 = T. The code ANT stands for which number?

12. A music box uses this code: 1 = L, 2 = E, 9 = G. What is the code for 921?

13. A robot arm uses this code: 4 = W, 0 = E, 9 = B. The code WEB stands for which number?

14. Using the code 0 = K, 1 = B, 2 = D, 3 = A, 4 = R, 5 = S, 6 = T, 7 = Y, 8 = F, 9 = L, select ALL the numbers below whose code would begin with the letter 'T'. A) 612 B) 348 C) 489 D) 673 E) 256

15. A jewellery box uses this code: 5 = S, 6 = I, 7 = T. The code SIT stands for which number?

Standard

20 questions

1. A code is formed by doubling a number and then adding 3. What is the code for 6?

2. A code is formed by multiplying a number by 2 and then subtracting 4. The code for a certain number is 18. What was the original number?

3. A code is formed by doubling a number and then subtracting 5. What is the code for 9? A) 11 B) 12 C) 13 D) 14

4. A code is formed by adding 3 to a number and then doubling the result. What is the code for 8?

5. A code is formed by halving a number and then adding 4. The code for a number is 10. What was the original number?

6. A code is formed by multiplying a number by 3 and then adding 4. The code for a number is 25. What was the original number? A) 6 B) 7 C) 8 D) 9

7. In a code, 2 → 9 and 4 → 17. The same rule is used for every number. What is the code for 6?

8. In a code, 3 → 13 and 6 → 25. The same rule applies to every number. If the code for a certain number is 17, what is that number?

9. A code is formed by multiplying a number by 4 and then adding 2. Select ALL the numbers below whose code would be greater than 30. A) 5 B) 6 C) 7 D) 8 E) 9

10. In a code, 4 → 19 and 7 → 31. Using the same rule, what is the code for 10? A) 41 B) 42 C) 43 D) 44 E) 45

11. A code is formed by subtracting 2 from a number and then multiplying the result by 4. What is the code for 7?

12. A code is formed by multiplying a number by 6 and then adding 2. The code for a number is 38. What was the original number?

13. A code is formed by subtracting 3 from a number and then multiplying by 5. What is the code for 8? A) 20 B) 22 C) 25 D) 28 E) 30

14. In a code, 3 → 10 and 5 → 16. What is the code for 9, using the same rule?

15. In a code, 4 → 11 and 9 → 26. If the code for a number is 20, what is that number, using the same rule?

16. A code is formed by doubling a number and then adding 7. What is the code for 11?

17. A code is formed by multiplying a number by 2 and then subtracting 6. The code for a number is 14. What was the original number?

18. A code is formed by adding 5 to a number and then doubling the result. What is the code for 7? A) 22 B) 24 C) 26 D) 20

19. A code is formed by halving a number and then adding 9. The code for a number is 15. What was the original number?

20. In a code, 2 → 12 and 5 → 24. Using the same rule, what is the code for 8?

Hard

5 questions

1. In a code, every digit has 4 added to it, wrapping back to 0 after 9. What is the code for 273?

2. A code adds 4 to every digit, wrapping after 9. If a number's code is 715, what was the number before coding?

3. Add 6 to every digit (wrap after 9) to make a code. Which of these is the code for 385? A) 941 B) 931 C) 951 D) 849 E) 942

4. In a code, the digits of a number are simply reversed. What is the code for 629?

5. Add 5 to every digit (wrap after 9) to make a code. Select ALL the numbers below whose code would include a 0 digit. A) 157 B) 246 C) 358 D) 503 E) 629

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